

# Brent Pappas

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GitHub: <https://www.github.com/PappasBrent>

## EDUCATION

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**PhD.** Computer Science. GPA: 3.983. University of Central Florida. Orlando, Florida. August 2021 - Present.

**Master of Science.** Computer Science. GPA: 3.983. University of Central Florida. Orlando, Florida. August 2021 - December 2025.

**Bachelor of Science.** Computer Science. GPA: 3.849. University of Central Florida. Orlando, Florida. August 2017 - May 2021.

## TEACHING

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| Fall 2026          | <b>Instructor of Record.</b> COP-3402 Systems Software. UCF. <a href="#">Course homepage</a> . <a href="#">Lecture Recordings</a> .                                                                                                                                                                                                                |
| Summer 2026        | <b>Instructor of Record.</b> COP-3402 Systems Software. UCF. <a href="#">Course homepage</a> . <a href="#">Lecture Recordings</a> . Mean average teaching evaluation score: <b>4.72/5</b> . 81/107 enrolled students responded. Department mean was 4.29, college mean 4.23. <a href="#">Student ratings</a> . <a href="#">Student comments</a> .  |
| Spring 2026        | <b>Instructor of Record.</b> COP-3402 Systems Software. UCF. <a href="#">Course homepage</a> . <a href="#">Lecture Recordings</a> . Mean average teaching evaluation score: <b>4.62/5</b> . 164/250 enrolled students responded. Department mean was 4.24, college mean 4.21. <a href="#">Student ratings</a> . <a href="#">Student comments</a> . |
| Fall 2025          | <b>Graduate Teaching Assistant.</b> COP-3402 Systems Software. UCF. Mean average overall teaching evaluation score: <b>4.93/5</b> . 15 students responded. <a href="#">Student perception of instruction</a> .                                                                                                                                     |
| Spring 2025        | <b>Graduate Teaching Assistant.</b> COP-3402 Systems Software. UCF.                                                                                                                                                                                                                                                                                |
| Spring 2021        | <b>Peer Tutor Mentor.</b> COP-3502 Computer Science 1 & COP-3330 Object Oriented Programming. UCF.                                                                                                                                                                                                                                                 |
| Spring & Fall 2020 | <b>Peer Tutor.</b> COP-3502 Computer Science 1 & COP-3330 Object Oriented Programming. UCF.                                                                                                                                                                                                                                                        |
| Fall 2019          | <b>Peer Tutor.</b> COP-3502 Computer Science 1. UCF.                                                                                                                                                                                                                                                                                               |

## HONORS AND AWARDS

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| 2026        | <b>GTA Excellence Award (University level).</b> For providing exceptional teaching assistance to the course Systems Software. Awarded to 1 out of 800+ graduate teaching assistants. <a href="#">Award letter</a> . |
| 2026        | <b>GTA Excellence Award (College level).</b> For providing exceptional teaching assistance to the course Systems Software. <a href="#">Award letter</a> .                                                           |
| 2026        | <b>CRA-WP 2026 Grad Cohort Workshop Invitee.</b> The CRA-WP arranged and paid for me to travel to the 2026 CRA-WP Grad Cohort Workshop in Seattle, Washington, because of my superior workshop application.         |
| 2024        | <b>UCF 3MT Finalist.</b> For effectively communicating my research to a general audience in a 3 minute talk. <a href="#">Video</a> . <a href="#">Certificate</a> .                                                  |
| 2024        | <b>Distinguished Artifact Reviewer.</b> For my attention to detail while reviewing artifacts for OOPLSA 2024. <a href="#">Link</a> .                                                                                |
| 2024        | <b>FCI Student Scholarship.</b> For my interest in interdisciplinary research, scholarship, and collaboration. \$5000 award.                                                                                        |
| 2021        | <b>Cum laude honors.</b> For having a cumulative undergraduate GPA greater than 3.8.                                                                                                                                |
| 2017 - 2021 | <b>Deans' list.</b> For continued undergraduate scholastic achievement.                                                                                                                                             |
| 2017        | <b>Burnett Honors Scholar.</b> For academic talent, motivation, intellectual curiosity and creativity.                                                                                                              |
| 2017        | <b>Provost Scholarship.</b> For outstanding academic performance in high school. Completely covered my undergraduate tuition fees at the University of Central Florida.                                             |

## PUBLICATIONS

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| 2026 | <b>Translation Tag Team: Formal Rules and LLMs Translate More Macros Together than Apart:</b> by Brent Pappas, Joseph Zalusky, Zachary Burkett, and Paul Gazzillo. ASE. <a href="#">Pre-print</a> . <a href="#">Artifact</a> . Badges: Available, functional, and reusable. |
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- 2026 Build Code is Still Code: Finding the Antidote for Pipeline Poisoning:** by Brent Pappas and Paul Gazzillo. ICSE NIER. Acceptance rate of 33.11% (50 / 151). [Pre-print](#). [Artifact](#). Badges: Available and reusable. [Presentation](#). [Slides](#).
- 2024 Semantic Analysis of Macro Usage for Portability:** by Brent Pappas and Paul Gazzillo. ICSE. Acceptance rate of 22.26% (234 / 1051). [Pre-print](#). [Artifact](#). Badges: Available and reusable. [Presentation](#). [Slides](#).
- 2023 Holy Macroni! A recipe for progressive language enhancement:** by Brent Pappas. Blog post for Trail of Bits. [Blog post](#). [Artifact](#).
- 2022 SugarC: Scalable Desugaring of Real-World Preprocessor Usage into Pure C:** by Zach Patterson, Zenong Zhang, Brent Pappas, Shiyi Wei, and Paul Gazzillo. ICSE. Acceptance rate of 28.51% (197 / 691). [Pre-print](#). [Artifact](#). Badges: Available and reusable. [Presentation](#).

## SERVICE

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- July 2026** [SPLASH/ISTTA 2026 Tool Demo Track Committee Member](#). Virtual.
- June 2026** **Camp Connect Lab Leader**. Orlando, Florida.
- March 2026** **NAC - NAPC Volunteer**. Orlando, Florida.
- October 2025** **UCF CS Grad Symposium 2025 Student Panelist**. Orlando, Florida.
- August 2025** **UCF GradLaunch PhD Panelist**. Orlando, Florida.
- August 2025** **UCF "Surviving Graduate School" Panelist**. Orlando, Florida.
- June 2025** **Camp Connect Lab Leader**. Orlando, Florida.
- May 2025** **UCF Graduate School Info Session PhD Panelist**. Orlando, Florida.
- April 2024** **UCF CECS Senior Design Judge**. Orlando, Florida.
- October 2022** **UCF CECS Career Kickoff Alumni Mentor**. Virtual.
- May 2024** Orlando, Florida.
- 2024** **OOPLSA 2024 Artifact Evaluation Committee Member**. Virtual.
- June 2023** **Camp Connect Lab Leader**. Orlando, Florida.
- Nov - Dec 2022** **EuroSys 2023 Shadow Program Committee Member**. Virtual.
- June 2021** **Camp Connect Lab Leader**. Orlando, Florida.

## PRESENTATIONS

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- 2026 Build Code is Still Code: Finding the Antidote for Pipeline Poisoning.** Presentation at the 2026 International Conference on Software Engineering in Rio de Janeiro, Brazil. [YouTube recording](#). [Slides](#).
- 2026 Curing Pipeline Poisoning.** 3 minute thesis on my research on securing build systems against pipeline poisoning with build phase isolation. Presented at the CRA 2026 Grad Cohort Workshop. [Recording link](#).
- 2026 Invited Talk - The PhD Life.** Guest lecture on graduate student life presented to Sonoma State University's 2026 Spring Colloquium, hosted by Dr. John Sullins. [Slides](#).
- 2025 Invited Talk - The PhD Life.** Guest lecture on graduate student life presented to Dr. Suzanne Rivoire's undergraduate course, Computing Professions, at Sonoma State University. [Slides](#).
- 2024 Semantic Analysis of Macro Usage for Portability (ICSE 2024).** Presentation at the 2024 International Conference on Software Engineering in Lisbon, Portugal. [Recording](#).
- 2023 Semantic Analysis of Macro Usage for Portability (CAE-R)** Virtual presentation for Center of Academic Excellence in Cyber Research (CAE-R). [Recording](#); [jump to 25:38](#).

## POSTERS

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- 2026** Curing Pipeline Poisoning with Build Phase Isolation. 2026 Academy of Science, Engineering & Medicine of Florida Annual Meeting. Acceptance rate of 55.56% (150 / 270).
- 2024** Holy Macroni! Finding Macro Usage Errors in the Linux Kernel. 2024 International Symposium on Secure and Private Execution Environment Design. [Link](#).

2022 Cpp2C: Transforming C Preprocessor Macros to C Language Constructs. 2022 University of Central Florida Student Scholar Symposium. [Link](#).

## CERTIFICATIONS

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[Completed higher education training, Preparing Tomorrow's Faculty](#)

[Level 1 College Readiness and Learning Assessment \(CRLA\) certified tutor.](#)

[Level 2 College Readiness and Learning Assessment \(CRLA\) certified tutor.](#)

UCF Student Enhancement Training (SET) program certified.

## WORKSHOPS

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**March 2026** 2026 CRA-WP Grad Cohort Workshop. Seattle, Washington.

**Oct - Nov 2025** 2025 CRA-WP Virtual Mentoring Series Workshops. Virtual.

**May 2025** 2025 CRA-E Career Landscape Workshop. Virtual.

**June 2022** Programming Language Mentoring Workshop (PLMW). San Diego, California.

## COMPLETE WORK EXPERIENCE

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**Jan 2026 - Present** **Graduate Teaching Associate (Instructor of Record) for COP-3402 Systems Software**  
University of Central Florida. Orlando, Florida.

**May 2024 - Dec 2025** **Graduate Research Assistant**  
University of Central Florida. Orlando, Florida.

**Aug 2023 - Apr 2024** **Graduate Teaching Assistant for COP-3402 Systems Software**  
University of Central Florida. Orlando, Florida.

**May 2023 - Aug 2023** **Software Engineering Intern**  
Trail of Bits. Remote.

**Jul 2021 - May 2023** **Graduate Research Assistant**  
University of Central Florida. Orlando, Florida.

**Sep 2019 - Jul 2021** **Quality Assurance Coordinator**  
SP+. Orlando, Florida.

**Jan 2021 - May 2021** **Peer Tutor Mentor**  
Student Academic Resource Center at the University of Central Florida. Orlando, Florida.

**Aug 2018 - May 2021** **Peer Tutor**  
Student Academic Resource Center at the University of Central Florida. Orlando, Florida.

## ADVISING

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### UNDERGRADUATE RESEARCHERS

**Aaron Burgess.** 2026 - Present. Human factors in software engineering.

**Thiago Airoidi.** 2026 - Present. Analysis and security of software development pipelines.

**Rasmin Ahmed.** 2026 - Present. Analysis and security of software development pipelines.

**Daniel Henriquez.** 2026 - Present. Analysis and security of software development pipelines.

**Brayden Coggin.** 2026 - Present. Analysis and security of software development pipelines.

**Thai Nguyen.** 2025 - 2026. Human factors in software engineering and analysis and security of software development pipelines.

**Nikola Krstajic.** 2025. Human factors in software engineering.

**Michael Johnson.** 2025 - Present. Human factors in software engineering and analysis and security of software development pipelines.

**Layla Le.** 2025. Human factors in software engineering.

**Kenneth Valladares.** 2025. Human factors in software engineering.

**Julian Velez.** 2025. Human factors in software engineering.

**Jorge Toledo.** 2025. Human factors in software engineering.

**Hasini Daliboyina.** 2025. Human factors in software engineering.

**David Gusmao.** 2025 - Present. Human factors in software engineering and analysis and security of software development pipelines.

**Anju Thomas.** 2025. Human factors in software engineering.

**Adam Betinsky.** 2025 - Present. Human factors in software engineering and analysis and security of software development pipelines.

**Ehigie Ekata.** 2024. Semantics-aware translation of C preprocessor.

**Zachary Burkett.** 2024. Semantics-aware translation of C preprocessor.

**Joseph Zalusky.** 2024. Semantics-aware translation of C preprocessor macros.

## PROJECTS

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**2026 Foreman.** University of Central Florida. A Python tool that enables developer to specify the commands file access permissions of phase of a project's build system, to protect these phases against pipeline poisoning attacks. Foreman is the tool associated with 2026 ICSE NIER paper, *Build Code is Still Code: Finding the Antidote for Pipeline Poisoning*.

**2024 Macroni.** Trail of Bits. A C static analysis tool that combines Trail of Bits' tools PASTA and VAST to lower C preprocessor macros down to MLIR. [GitHub](#).

**2024 Maki.** University of Central Florida. A C++ Clang plugin that analyzes hows C preprocessor macros affect the C AST. Maki provides language porting tools information they can use to port macros to target languages in a way that preserves not just macro behavior, but also macro abstractions. Maki is the tool associated with 2024 ICSE paper, *Semantic Analysis of Macro Usage for Portability*, for which I am the lead-author. [GitHub](#).

## SKILLS SUMMARY

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**Languages:** Python3, C/C++, Bash, and HTML/CSS/JavaScript

**Tools:** Git, GitHub, and LLVM/Clang

**Platforms:** Linux, Web, and Windows

**Soft Skills:** Leadership, Writing, Communication, and Time management